

Deep Learning Assignment 5: Vehicle Detection

1. Methodology & Augmentation

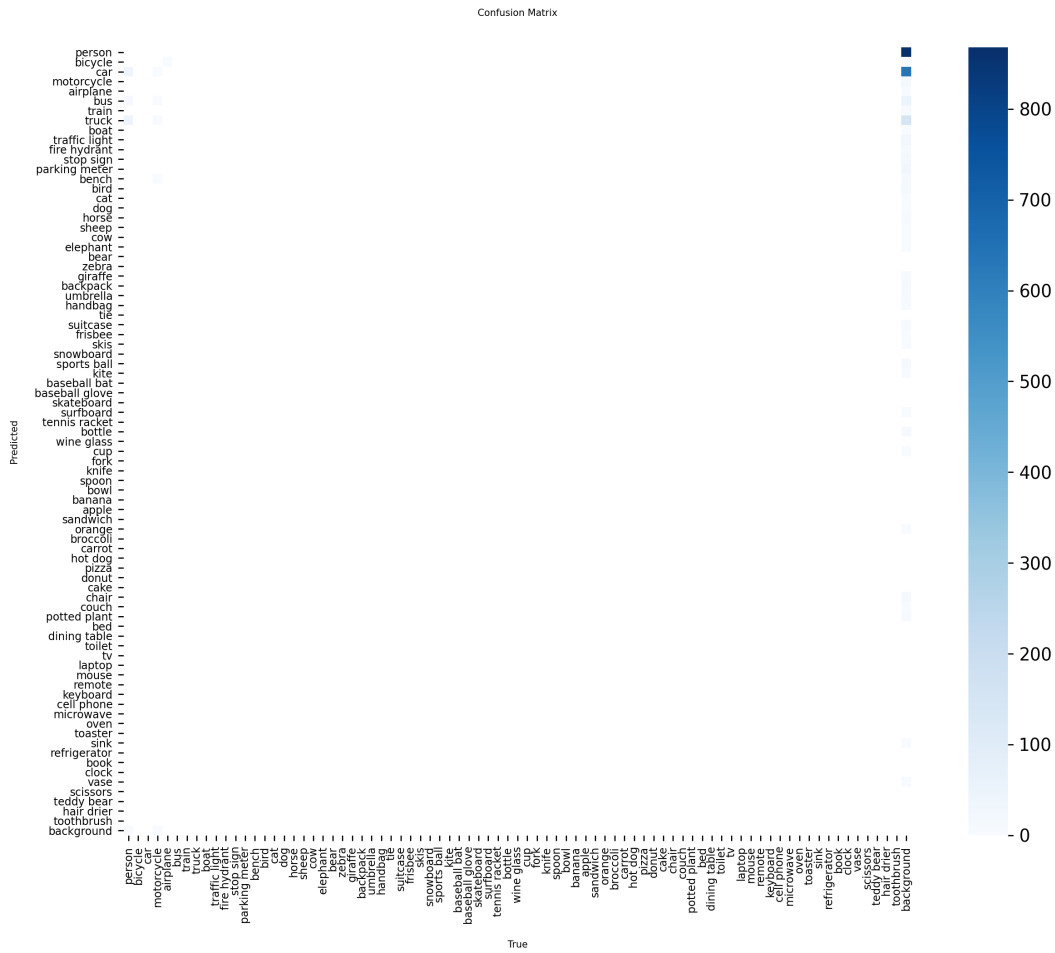
- Dataset: 100 images, split into 70 Train, 15 Val, 15 Test.
- Augmentation: Mosaic, flip, scale, and color jitter applied ONLY to training set.
- Models: YOLOv8s, YOLOv8m, YOLOv8l (Fine-tuned from COCO pretrained weights).

2. Class Distribution

- Car: 513 instances
- Truck: 22 instances
- Motorcycle: 5 instances
- Bus: 3 instances

Analysis: The dataset is imbalanced (e.g., 'Car' dominates). This may cause lower recall for minority classes like 'Bicycle' or 'Motorcycle'.

3. Confusion Matrix (Test Set)



4. Prediction Examples



5. Training Metrics (W&B Tracking)

All experiments were tracked using Weights & Biases.

Project: <https://wandb.ai/aliazar499-azar/dl-assign5-vehicle-detection>

